Name: _____ Period: _____

Projectile Launch Project: Planning Pages

Tier: Target / Extension **In class:** Projectile Launch Project (after the unit test)

About this page

- Use these pages with the Projectile Launch Project handout. They organize your thinking; your formal write-up goes in your lab document.

Key ideas

- Task 1 must use only a meter stick. No stopwatches.
- Task 2: you will know v_0, the launch height, g, and the launch angle. Every equation must use only those.
- Practice problem 3 uses a different setup on purpose. Use it to rehearse the method, not to copy numbers.

1. Assumptions. List at least three assumptions your model of the launch makes.

2. Task 1 plan. Sketch how you will set up the launcher to find v_0 with only a meter stick. Label every distance you will measure and say which variable it is.

3. Rehearsal problem. A launcher on a 1.20 m lab table fires at 4.5 m/s, 25° above the horizontal, onto the floor. Find:
a) the total time to fall b) the maximum height above the floor c) the horizontal final position

4. Sources of error. Name two measurements that could throw off your target placement and how you will reduce each one.

5. After launch day. Predicted landing: _____ Actual landing: _____ Percent difference: _____

What most likely explains the difference?

6. Percent difference practice. A group predicted a landing at 2.85 m and measured 2.71 m. Find the percent difference from the prediction.

7. Challenge: change the angle. Rework the rehearsal problem at 45° instead of 25°. Does the ball land farther? Explain why 45° is not automatically best when you launch from a height.

8. Challenge: measurement error. In the rehearsal problem, suppose you measured the height as 1.22 m instead of 1.20 m. How much does the predicted landing spot move?

9. Think it through. If the rehearsal launcher sat on the floor ($y_i = 0$) instead of the table, would the ball land closer or farther? Explain without calculating.